

2024 Bi H1 Q12

Section: Metabolism and Survival

Topic: Respiration in Yeast – Control Experiment

Question Summary:

An investigation studied the effect of an enzyme inhibitor on respiration rate in yeast. Each flask contained yeast, glucose, and inhibitor at different concentrations. The question asks what the suitable control flask should contain.

Worked Solution:

- A control experiment must be identical to the test setups except for the independent variable (the inhibitor concentration).
- To control for the inhibitor variable, the control flask should contain the same total volume of liquid but with water replacing the inhibitor.
- This ensures that any change in respiration rate (measured by CO₂ production) is due to the inhibitor, not volume or composition changes.

Therefore, the control flask should contain 5 cm³ yeast solution, 5 cm³ glucose solution, and 1 cm³ water.

Final Answer: **B** (5 cm³ yeast solution, 5 cm³ glucose solution, and 1 cm³ water).

Revision Tips:

- A control contains all components except the independent variable.

- Replacing the inhibitor with water ensures equal total volume and conditions.
- Respiration rate can be measured via CO₂ output or oxygen uptake.